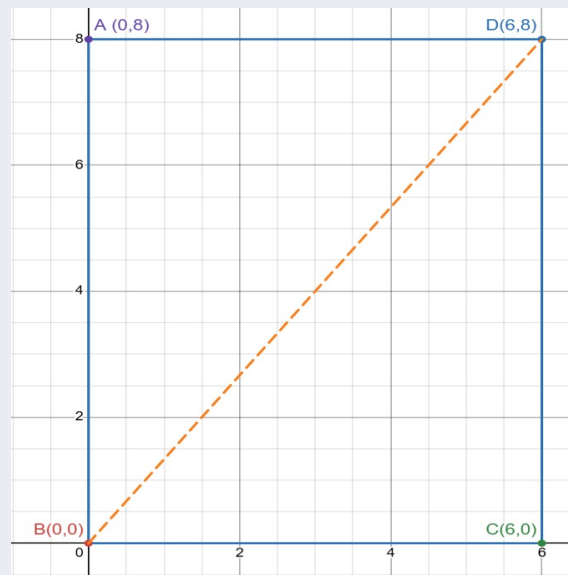


WEEKLY PROBLEM

The "Phantom Fold" Challenge

EXERCISE.

Consider a perfectly flat rectangular piece of paper ABCD where the side lengths are $AB = 8$ units and $BC = 6$ units.



1. The 2D crease

The paper is lying flat on a desk. You fold the paper such that vertex C perfectly coincides with vertex A. This fold creates a straight line crease across the paper. Calculate the exact length of this crease line.

2. The 3D Transformation

You unfold the paper back to its original flat rectangular state ABCD. Next, you make a new fold along the diagonal BD. You fold it such that the two triangular halves ($\triangle ABD$ and $\triangle CBD$) form a dihedral angle of 90° (meaning the two planes containing the triangles are exactly perpendicular to each other in 3D space). Calculate the shortest straight-line distance in 3D space between the vertices A and C.

3. The Enclosure

Using the exact 3D folded configuration from part 2 (where the planes are perpendicular), find the radius of the **smallest possible sphere** that can completely enclose all the four vertices (A, B, C and D) of the folded paper.



Top Submissions for Problem (WP2608):

1. **Aman Raj Kumar** - MC25B1005

2. **Bhupati Shri Akshit** - MC24B1010